out-going); this encodes the fact that in a 2-3 tree there are only 2-nodes and 3-nodes.
2) All paths from the root to a leaf have the same number of black links.
Equivalently, nodes can be marked as red or black (here, we used red and white for clarity), and there can't be two consecutive red nodes in any path.
Together, these constraints guarantees that the longest possible path in a red-black BST, alternating red and black links, can at most be twice as long as the shortest possible path in the tree, containing only black links. In turn, this guarantees that the height of the tree is logarithmic.
These invariants are maintained by appropriately using rotations, after insertions and deletions.

Figure 3.14 illustrates the process: the right branch of the tree is not balanced, and by updating the second-to-last-level node, we can force a right rotation that will bring it up one level, re-balancing its subtree (and, in this simple example, the whole tree).

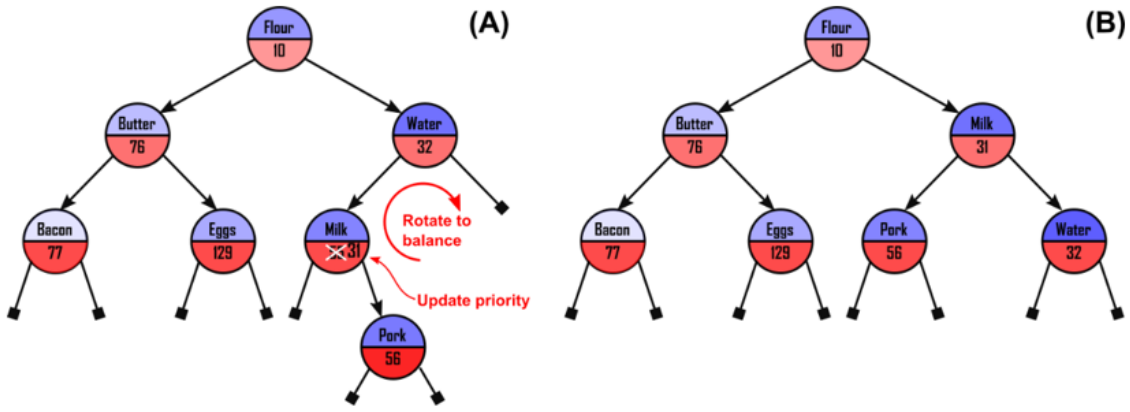


Figure 3.14 Rebalancing a treap by carefully updating its priorities. If we change the priority of the node with key "Milk" to a value smaller than its parent (but larger than